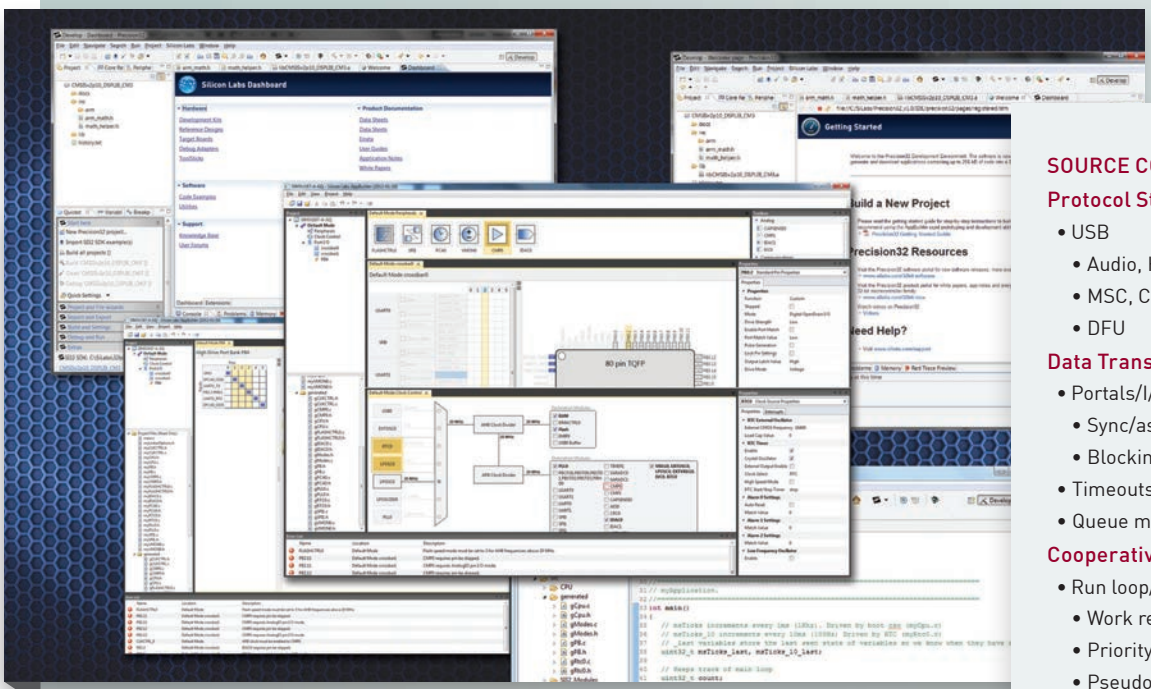
MCU SOLUTIONS

Precision32™ 32-bit Software Tools

www.silabs.com/32bit-software



ENGINEERED TO MAKE YOUR JOB EASIER

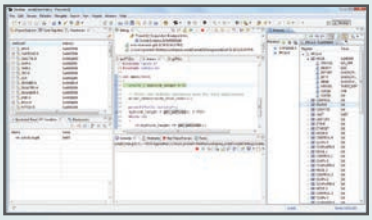
The Precision32™ SiM3U1xx and SiM3C1xx MCUs provide a highly integrated and flexible architecture and a rich peripheral set. To ease design, Silicon Labs offers intuitive software tools available free to developers, including an Eclipse-based integrated development environment (IDE) with full compiler, debugger and dashboard, and the AppBuilder rapid prototyping and code generation utility.

The free Precision32 Development Suite includes the following components, which are described in detail below:

- **Precision32 Integrated Development Environment (IDE)**
- **Precision32 AppBuilder Rapid Prototyping Utility**
- **Si32 SDK (HAL, Software Libraries and Development Kit Utilities)**

Precision32 Integrated Development Environment

Silicon Labs' Eclipse-based IDE includes a full compiler, debugger and an online dashboard for application-critical information such as a software libraries with example code, data sheets, schematics, PCB footprints, app notes, active version tracking and automatic updates. These tools are available for free, without code size or time restrictions.



SOURCE CODE AND LIBRARIES:

Protocol Stacks

- USB
  - Audio, HID
  - MSC, CDC
  - DFU

Data Transfers

- Portals/I/O objects
  - Sync/async
  - Blocking/non-blocking
- Timeouts/cancellation
- Queue management

Cooperative Multi-tasking

- Run loop/dispatcher
  - Work request
  - Priority queue
  - Pseudo-thread

Interface Abstraction

- Serial
  - UART, USART, I²C
  - I²S, SPI
  - USB

Infrastructure Support

- Memory Allocation
  - Static
  - Dynamic one-time
  - Dynamic heap-based
- Data Collections
  - Buffers
  - Pools
  - Queues
- Timing Generation
  - Start/stop/interval
  - Repeat/one-shot
- Run-time Debug Logging
  - Start/stop
  - Print/trace/warning
- Error handling
  - Assert
  - Throw/catch

FREE, EASY-TO-USE SOFTWARE TOOLS

MORE INFORMATION AND DOCUMENTATION DOWNLOADS AT: www.silabs.com/32bit-software

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Engineering for a mixed-signal world.

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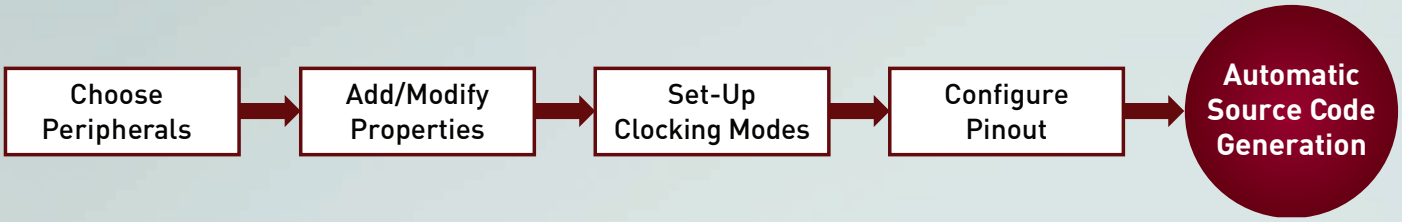
Precision32 AppBuilder Rapid Prototyping Utility

Silicon Labs' Precision32 AppBuilder graphical rapid prototyping and code generation utility helps designers get up and running quickly. Designers can easily set-up and adjust pin configuration, map peripherals, configure clocking trees and generate source code all without wading through programming register bits in the data sheet. A sophisticated error checking engine validates peripheral properties and pin assignments, and provides real-time feedback when conflicts are detected.

Last minute changes... No problem. New peripherals can be easily added with simple firmware modifications so developers won't have to work weekends to try to accommodate new features.

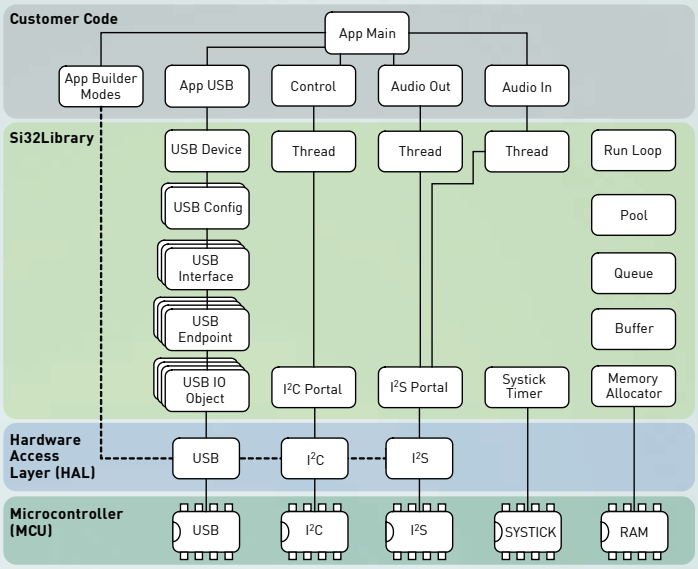


Extensive Software Library

Silicon Labs Precision32 microcontroller library provides building blocks for creating new stacks, allowing developers to focus on their end application instead of spending time on MCU specific details.

Made with simplicity in mind, the Si32Library easily hooks into the main application via function calls. The Si32Library is comprised of flexible, scalable and portable source modules enabling core functionality for error handling, debug logging, memory allocation, buffer management, data collections, data transfers, event notification, multitasking, USB, tool chain portability, RTOS compatibility, and more. Portable objects connect directly to Silicon Labs HAL (hardware access layer), which are easily configured using AppBuilder.

The Si32Library is delivered as source code as part of the Precision32 Development Suite installation. Customers can modify and adapt it to tune performance and memory as required by their application. A simple object-based model allows for reuse and modularity without the added complexity of C++.



ARM® Ecosystem

The Precision32 family also enables developers to take advantage of the huge ecosystem of software and development tools optimized for the ARM® architecture including Keil, IAR and Micrium tool chains and RTOS and protocol stacks. To accelerate development and code porting from competitive MCU platforms, Silicon Labs offers Cortex-M3 microcontroller software interface standard (CMSIS) compatible code, an ARM-specified software interface for core peripherals.

Download the Precision32 Software Development Tools Today

Silicon Labs provides both a web and offline installer to download the Precision32 Development Environment. Both options are provided at no cost to you and include the full Precision32 Development Suite, including the IDE, AppBuilder, hardware access layer, software libraries and development kit utilities. Download today!

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